Sports Med (2017) 47:975–986
DOI 10.1007/s40279-016-0634-6

SYSTEMATIC REVIEW

Effect of Plyometric Training on Vertical Jump Performance in Female Athletes: A Systematic Review and Meta-Analysis

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Published online: 4 October 2016

Abstract

Background Plyometric training is an effective method to prevent knee injuries in female athletes; however, the effects of plyometric training on jump performance in female athletes is unclear.

Objective The aim of this systematic review and meta-analysis was to determine the effectiveness of plyometric training on vertical jump (VJ) performance of amateur, collegiate and elite female athletes.

Methods Six electronic databases were searched (PubMed, MEDLINE, ERIC, Google Scholar, SCIndex and ScienceDirect). The included studies were coded for the following criteria: training status, training modality and type of outcome measures. The methodological quality of each study was assessed using the physiotherapy evidence database (PEDro) scale. The effects of plyometric training on VJ performance were based on the following standardised pre–post testing effect size (ES) thresholds: trivial (<0.20), small (0.21–0.60), moderate (0.61–1.20), large (1.21–2.00), very large (2.01–4.00) and extremely large (>4.00).

Results A total of 16 studies met the inclusion criteria. The meta-analysis revealed that plyometric training had a most likely moderate effect on countermovement jump (CMJ) height performance (ES = 1.09; 95 % confidence interval [CI] 0.57–1.61; I^2 = 75.60 %). Plyometric training interventions of less than 10 weeks in duration had a most likely small effect on CMJ height performance (ES = 0.58; 95 % CI 0.25–0.91). In contrast, plyometric training durations greater than 10 weeks had a most likely large effect on CMJ height (ES = 1.87; 95 % CI 0.73–3.01). The effect of plyometric training on concentric-only squat jump (SJ) height was likely small (ES = 0.44; 95 % CI −0.09 to 0.97). Similar effects were observed on SJ height after 6 weeks of plyometric training in amateur (ES = 0.35) and young (ES = 0.49) athletes, respectively. The effect of plyometric training on CMJ height with the arm swing was likely large (ES = 1.31; 95 % CI −0.04 to 2.65). The largest plyometric training effects were observed in drop jump (DJ) height performance (ES = 3.59; 95 % CI −3.04 to 10.23). Most likely extremely large plyometric training effects on DJ height performance (ES = 7.07; 95 % CI 4.71–9.43) were observed following 12 weeks of plyometric training. In contrast, a possibly small positive training effect (ES = 0.30; 95 % CI −0.63 to 1.23) was observed following 6 weeks of plyometric training.

Conclusion Plyometric training is an effective form of training to improve VJ performance (e.g. CMJ, SJ and DJ) in female athletes. The benefits of plyometric training on VJ performance are greater for interventions of longer duration (≥10 weeks).

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Key Points

Plyometric training was associated with very large increases in drop jump (DJ) performance, large increases in countermovement jump (CMJ) performance and small increases in squat jump (SJ) performance.

Longer training durations (≥10 weeks) provide larger improvements in jump performance.

Plyometric training is an effective form of training to improve vertical jump performance (e.g. CMJ, SJ and DJ) in female athletes.

1 Introduction

Plyometrics is a well-known form of 'ballistic training', designed to improve jump performance capabilities [1]. In addition, plyometric training has been shown to be an effective method for improving strength [2], running economy [3], agility [4, 5] and sprint ability [5, 6]. Plyometric training has also been utilised to help prevent knee injuries [7, 8]. It has been shown that female athletes have a greater risk of knee injury than male athletes [9–14], due to a larger Q-angle, causing increased valgus as well as weakness in the surrounding musculature. Plyometric training may improve landing mechanics (reduce valgus stress and strain), improve eccentric muscle control, and increase knee flexion and hamstrings activity, which in turn reduce landing forces and minimise the risk of non-contact injuries [8, 15, 16]. Young developing female athletes (age 12–18 years) with minimal plyometric training experience appear to be the most responsive to plyometric training in terms of improving vertical jump (VJ) performance and reducing injury risk [17–19]. Historically, plyometric training was thought to be unsafe for youth athletes, as a certain level of strength (back squat ≥150 % of body mass) was recommended as a prerequisite for partaking in plyometric training [20]. However, this contention was refuted and revised based on the findings of emerging youth development research, and plyometric training is now deemed an effective method of improving a number of physical qualities in youth [21–23].

Previous studies have concluded that the effectiveness of plyometric training depends on the volume and frequency of training [24], and a number of subject characteristics, such as strength training level [25], sex [26, 27], age [22, 27] and sport [28]. Neural recruitment and movement pattern development in males and females is constantly changing through the stages of maturation and into adulthood. In comparison with females, males develop greater increases in power, strength and coordination with age that correlate with the stages of maturation [29, 30]. It has been observed that VJ height (VJH) performance improves during maturation in males, but not in females [29, 30]. VJH is the primary criterion measure and the most reported variable to assess VJ ability. Jump performance ability is governed by an individual's ability to utilise the elastic and neural benefits of the stretch-shorten cycle (SSC) [26, 31], along with the force and rate of excursion of the activated musculature during contraction. Higher forces are associated with a shorter amortisation (transition) time from the eccentric phase to the concentric phase [32] and with greater energy stored in the series elastic component [33].

There are a number of reviews written on the benefits of plyometric training; however, the majority of these reviews focus on male participants. A meta-analytic review by Markovic [34] confirmed the effectiveness of plyometric training on VJ performance of males and females using pooled statistical methods; however, the majority of the included studies were conducted on male participants (83 % males vs. 17 % females). De Villarreal et al. [24] demonstrated that males (effect size [ES] = 0.80) had greater gains in VJH than females (ES = 0.50) following plyometric training; they conducted a meta-analysis of 56 studies, comprised of 47 male and nine female studies. Similar to the above findings, de Villarreal et al. [28] conducted a meta-analysis of 15 plyometric training studies (11 male; four female) and found slightly larger strength increases in males (ES = 1.01) than in females (ES = 0.72); however, both sexes had moderate gains in strength following resistance training. The large differences in sample size between male (n = 58) and female (n = 13) studies analysed in these meta-analyses may have confounded the results. Based on current reviews, the effects of plyometric training on VJH performance in females remain unclear; therefore, further analysis is necessary to determine the true effects of plyometric training VJH in female athletes. In addition, the number of plyometric training interventions conducted on female athletes has increased in recent years. Therefore, the aim of this systematic review and meta-analysis was to determine the effectiveness of plyometric training interventions on VJH performance in amateur, collegiate and elite female athletes.

2 Methods

2.1 Search Strategy

An extensive electronic database search was performed using PubMed, MEDLINE, ERIC, Google Scholar, SCIndex and ScienceDirect to identify relevant articles that investigated the effects of plyometric training on VJH performance of female athletes. A manual search was performed using a combination of the following key terms: plyometric training, ballistic training, jump training, plyometric exercise, vertical jump, vertical leap, squat jump, countermovement jump, drop jump, depth jump, effects, improvement, increases, performance, female and women. The reference lists of each included article were also scanned to identify additional relevant studies. The database search was limited to peer-reviewed journal articles published in English between 1996 and 2016.

The electronic searches, identification, screening and data extraction were conducted independently by two reviewers (ES and VR). Disagreements were resolved by a third reviewer (ZM). This systematic review and meta-analysis was undertaken in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [35].

2.2 Inclusion Criteria

2.2.1 Type of Study

This review included randomised and non-randomised controlled trials written in English and published within the last 20 years.

2.2.2 Type of Participants

The included participants were healthy amateur, collegiate and/or elite female athletes of any age (no restriction).

2.2.3 Type of Interventions

To be included in the meta-analysis, plyometric training interventions were required to be a minimum of 4 weeks in duration and include a control group. The control groups either participated in a general sports programme without plyometric training or did not exercise.

2.2.4 Type of Outcome Measure

The included articles were required to report VJH as the outcome measure during one or more of the following movement patterns: squat jump (SJ), countermovement jump (CMJ) with and without arm swing and drop jump (DJ).

2.3 Exclusion Criteria

The exclusion criteria were as follows: (1) studies using male participants; (2) studies written in languages other than English; (3) studies without a control group; (4) studies with a combination of plyometric and weight training in order to avoid effects of combined training; (5) uncontrolled and cross-sectional studies; and (6) studies where the results were not presented separately for males and females.

2.4 Data Extraction

Cochrane Consumers and Communication Review Group's data extraction protocol was used to extract participant information, including health status, sex, sample size, description of the intervention (type of exercises, intensity, duration and frequency), study design and study outcomes. This was undertaken by one author (VR), while a second author (ES) checked the extracted data for accuracy and completeness. The quality assessment was conducted by one author (ES). Disagreements were resolved by consensus or by a third reviewer. Reviewers were not blinded to authors, institutions or manuscript journals.

2.5 Quality Assessment

The Physiotherapy Evidence Database (PEDro) scale was used to assess methodological quality of the included studies [36]. Quality assessment was interpreted using the following 10-point scale: $\leq$3 points was considered poor quality, 4–5 points as moderate quality and 6–10 points as high quality. The PEDro scale consists of 11 items designed for rating methodological quality. Each satisfied item contributes 1 point to the overall PEDro score (range 0–10 points). Item 1 was not included as part of the study quality rating as it pertains to external validity.

2.6 Statistical Analyses

Meta-analyses were performed using Comprehensive Meta-analysis software (Version 2, Biostat Inc., Englewood, NJ, USA). The mean differences and 95 % confidence intervals (CIs) were calculated for the included studies. The I^2 measure of inconsistency was used to examine between-study variability; values of 25, 50 and 75 % represent low, moderate and high statistical heterogeneity, respectively [37]. A random effects meta-analysis was conducted to determine the pooled effect of plyometric training on CMJ (with and without arm swing), SJ and DJ performance. In addition, training intervention duration ($\leq$10 and >10 weeks) was used as moderator variable for CMJ. ESs were calculated when raw mean values, standard deviations (SDs), and sample sizes (n) were available, using the following formula (Eq. 1:

$$\mathrm{ES} = \frac{\text{Raw mean change}_1 - \text{Raw mean change}_2}{\mathrm{SD}_{Post-Pooled}} \tag{1}$$

$SD_{Post\text{-}Pooled}$ was calculated using the following formula (Eq. 2):

$$SD_{Post-Pooled} = \sqrt{\frac{(n_1 - 1)SD_1^2 + (n_2 - 1)SD_2^2}{n_1 + n_2 - 2}} \quad (2)$$

The chance of the true effect being trivial, beneficial or harmful was interpreted using the following scale: 25–75 % (possibly); 75–95 % (likely); 95–99.5 % (very likely); and 99.5 % (most likely) [38]. The magnitude of the plyometric training effects on VJH were interpreted using the following criteria: trivial (<0.20), small (0.21–0.60), moderate (0.61–1.20), large (1.21–2.00), very large (2.01–4.00) and extremely large (>4.00) changes [38].

3 Results

3.1 Study Selection

A total of 533 articles were identified from the database search with an additional ten articles identified through reference lists. Following the removal of duplicates and the elimination of articles based on title and abstract screening, 85 studies remained. An evaluation of the remaining 85 studies was conducted independently by two researchers. Following the final screening process, 16 studies were included in the systematic review and meta-analysis (Table 1). Details of the study selection process has been presented in Fig. 1.

3.2 Methodological Quality

Overall, the included studies were of moderate quality, with a mean PEDro score of 5.25 (Table 2). Only one study [39] scored positive for blinding of therapists and assessors. Two studies satisfied the item for blinding of participants [39, 40]. Low scores were also found for 'intention to treat' (all studies failed to satisfy this item) and 'allocation concealed' (two studies satisfied this item). All of the included studies received points for the following items: baseline indicator, measures of at least one key outcome, statistical comparison between groups and point measure. Fourteen studies received moderate quality scores [17, 18, 41–52]; only two studies received high quality ratings [39, 40].

3.3 Study Characteristics

All studies that met the inclusion criteria were published in English between 2004 and 2015. The pooled sample size of the 16 studies was 431, and the typical sample size of the individual studies ranged from eight to 20 subjects per group. The intervention durations of the included studies were as follows: 6 weeks ($n = 7$ studies) [39–41, 43, 48, 51, 52], 8 weeks ($n = 3$ studies) [18, 44, 50], 9 weeks ($n = 1$ study) [45], 10 weeks ($n = 1$ study) [49], 12 weeks ($n = 3$ studies) [17, 42, 47] and 26 weeks ($n = 1$ study) [46]. Fifteen studies investigated the effects of land-based plyometric training; one examined the effect of aquatic plyometric training [48]. The most common training frequency was two to three sessions per week (mean $\pm$ SD 2.06 $\pm$ 0.56). Plyometric training sessions lasted between 20 and 120 min (mean $\pm$ SD 53.24 $\pm$ 37.07). However, one study [46] implemented a plyometric training intervention that consisted of 30 jumps per week (10 jumps per day) for a duration of 6 months. The mean ($\pm$SD) number of jumps completed per session and per week were as follows: 91.09 $\pm$ 50.21 and 168.25 $\pm$ 89.44, respectively. The total number of jumps completed across the respective plyometric training interventions ranged from 360 to 3320 jumps (mean $\pm$ SD 1543.8 $\pm$ 997.81).

3.4 Study Outcomes

3.4.1 *Countermovement Jump Without Arm Swing*

The effect of plyometric training on CMJ height was most likely moderate (ES = 1.09; 95 % CI 0.57–1.61; $I^2 = 75.60$ %). The effects of plyometric training interventions of less than 10 weeks in duration were most likely small (ES = 0.58; 95 % CI 0.25–0.91); in contrast, the effects of plyometric training interventions greater than 10 weeks in duration were most likely large (ES = 1.87; 95 % CI 0.73–3.01). The measures of heterogeneity for the plyometric training intervention durations of less than 10 weeks and more than 10 weeks were equal to zero ($I^2 = 0$ %) and high ($I^2 = 87.23$ %), respectively (Fig. 2).

3.4.2 *Countermovement Jump with Arm Swing*

Similar trends were observed during the CMJ with arm swing, as the plyometric training interventions had likely large effect on CMJ height performance (ES = 1.31; 95 % CI –0.04 to 2.65; Fig. 3). The heterogeneity for plyometric training on the CMJ with the arm swing was high ($I^2 = 92.68$ %). Most likely extremely large training effects (ES = 5.10; 95 % CI 4.05–6.14) were observed in amateur female athletes with minimal plyometric training exposure (duration = 8 weeks, frequency = two sessions per week; 188 jumps per week). Conversely, the effect of plyometric training in elite female athletes was likely trivial (ES = 0.19; 95 % CI –0.93 to 1.31)

Table 1 Systematic review and characteristics of included studies selected for meta-analysis and relevant outcomes

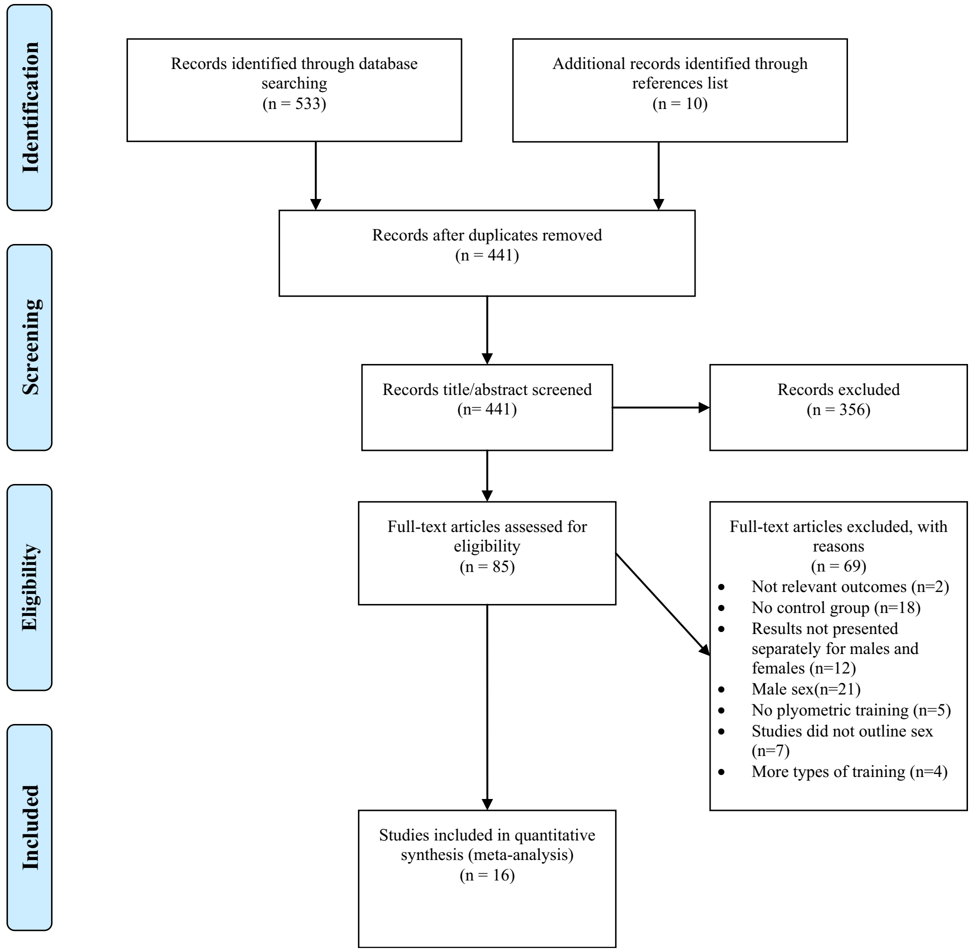
Study	Population			Training programme					Outcomes			
	Level	Age (years) [mean ± SD]	Duration (weeks)	Sample size (n)	Intensity	Days/week	Session (min)	Jumps completed	CMJ (%)	SJ (%)	CMJA (%)	DJ (%)
Attene et al. [52]	Basketball players	14.9 ± 0.9	6	TG (18) CG (18)	Medium and high	2	20	COMB 560	TG: 11.3 % ↑* CG: 4.6 % ↑	TG: 15.4 % ↑* CG: 7.5 % ↑		
Ramírez-Campillo et al. [39]	Amateur soccer players	TG 22.9 ± 1.7 CG 22.5 ± 2.1	6	TG (10) CG (10)	High	2	120	COMB 1215	TG: 4.4 % ↑* CG: 0.5 % ↑	TG: 5.1 % ↑* CG: 0.7 % ↓		
Ramírez-Campillo et al. [41]	Elite runners	22.1 ± 2.7	6	TG (8) CG (5)	High	2	30	DJT 360			TG: 6.5 % ↑* CG: 4.3 % ↑	
Ramírez-Campillo et al. [40]	Collegiate soccer players	TG 22.4 ± 2.4 CG 20.5 ± 2.5	6	TG (19) CG (19)	High	2	120	COMB 1095	TG:10.7 % ↑* CG: 0.5 % ↑		TG: 8.3 % ↑* CG:1.1 % ↓	
Campo et al. [42]	Collegiate soccer players	22.8 ± 2.1	12	TG (10) CG (10)	High	3	40–60	COMB 3320	TG: 12.9 % ↑* CG: 2.7 % ↓			TG: 18.1 % ↑* CG: 7 % ↓*
Chimera et al. [43]	Collegiate soccer and hockey players	18–22	6	TG (9) CG (9)	High	2	20–30	COMB 1970				TG: 5.6 % ↑ CG: 1.8 % ↑
Irmischer et al. [45]	College students	TG 24.0 ± 5.0 CG 24.3 ± 4.3	9	TG (14) CG (14)	Low	2	20	COMB 698	TG: 14.3 % ↑ CG: 8.6 % ↑			
Kato et al. [46]	College students	TG 20.5 ± 0.6 CG 20.9 ± 0.8	26	TG (18) CG (18)	Low	3	2	CMJA 780			TG: 9.1 % ↑* CG: 3 % ↑*	
Makaruk et al. [47]	College students	TG 20.6 ± 1.3 CG 21.0 ± 1.9	12	TG (16) CG (15)	Low, medium and high	2	40–45	COMB Unilateral 3244	TG: 17.4 % ↑* CG: 4.8 % ↑			
Makaruk et al. [47]	College students	TG 20.9 ± 1.7 CG 21.0 ± 1.9	12	TG (18) CG (15)	Low, medium and high	2	40–45	COMB Bilateral 3244	TG: 22.8 % ↑* CG: 4.8 % ↑			
Martel et al. [48]	Highschool volleyball players	TG 15 ± 1 CG 14 ± 1	6	TG (10) CG (9)	Low, medium and high	2	45	ACOMB			TG: 11.1 % ↑* CG: 4 % ↑*	

Table 1 continued

Study	Population			Training programme					Outcomes			
	Level	Age (years) [mean ± SD]	Duration (weeks)	Sample size (n)	Intensity	Days/week	Session (min)	Jumps completed	CMJ (%)	SJ (%)	CMJA (%)	DJ (%)
Ozbar et al. [50]	Collegiate soccer players	TG 18.3 ± 2.6 CG 18.0 ± 2.0	8	TG (9) CG (9)	Low, medium and high	1	60	COMB 1210	TG: 17.6 % ↑* CG: 7.1 % ↑*			
Ozbar [49]	Collegiate soccer players	TG 19.4 ± 1.6 CG 19.1 ± 1.7	10	TG (10) CG (10)	Low, medium and high	2	40	COMB 800–930	TG: 21.2 % ↑* CG: 6.5 % ↑*			
Pereira et al. [18]	Highschool volleyball players	TG 14.0 ± 0.0 CG 13.8 ± 0.4	8	TG (10) CG (10)	Medium and high	2	20	COMB 1188	TG: 20.2 % ↑* CG: 3.2 % ↑			
Rubley et al. [17]	Local soccer league	13.4 ± 0.5	12	TG (10) CG (6)	Low	1	90–120	COMB 1920			TG: 18.6 % ↑* CG: 6.6 % ↓	
Usman and Shenoy [44]	Collegiate volleyball players	18–22	8	TG (30) CG (30)	Low, medium and high	2	90–120	COMB 1488			TG: 18.7 % ↑* CG: 0.6 % ↑	
Vescovi et al. [51]	College students	TG 20.3 ± 1.2 CG 19.9 ± 1.6	6	TG (10) CG (10)	Low, medium and high	3	45–60	COMB	TG: 3.1 % ↑ CG: 0.6 % ↓			

ACOMB combination of various jump exercises in water, *CG* control group, *CMJ* countermovement jump, CMJA countermovement jump with the arms swing, *COMB* combination of various jump exercises, *DJ* drop jump, *DJT* drop jump exercise, *SD* standard deviation, *SJ* squat jump, *TG* treated group, ↑* indicates significant increase, ↑ indicates increase

Fig. 1 Flow chart diagram of the study selection



(duration = 6 weeks, frequency = two sessions per week; 60 jumps per week).

3.4.3 Squat Jump

The effect of plyometric training on SJ height performance was likely small (ES = 0.44; 95 % CI −0.09 to 0.97). Heterogeneity of the effect of plyometric training on SJ height was equal to zero ($I^2 = 0$ %). Similar effects following 6 weeks of plyometric training were observed in amateur (ES = 0.35; 95 % CI −0.54 to 1.23) and young athletes (ES = 0.49; 95 % CI −0.17 to 1.15).

3.4.4 Drop Jump

The effects of plyometric training on DJ height performance was likely very large (ES = 3.59; 95 % CI −3.04 to 10.23). The heterogeneity of effects of plyometric training across the DJ studies was high ($I^2 = 96.35$ %). The highest effect on DJ height (ES = 7.07; 95 % CI 4.71–9.43) was observed following 12 weeks of plyometric training (frequency = three sessions per week; 277 jumps per week); the effect was most likely extremely large. In contrast, a possibly small effect (ES = 0.30; 95 % CI −0.63 to 1.23) was observed following 6 weeks of plyometric training (frequency = two sessions per week; 328 jumps per week).

4 Discussion

The results of this systematic review and meta-analysis confirm the effectiveness of plyometric training on VJ performance in female athletes. The plyometric training effects on SJ, CMJ and DJ performance were small, moderate to large and very large, respectively. Overall, the effects of plyometric training interventions of longer than 10 weeks in duration (ES = 1.87) were much higher than for interventions of less than 10 weeks (ES = 0.58). This indicates that the benefits of plyometric training on VJ performance are greater for interventions of longer duration (≥10 weeks). Furthermore, this review revealed that short plyometric training interventions (6 weeks) only provide a small training effect on jump performance in female athletes [39–41, 48, 52]. Additionally, two studies

Table 2 Quality assessment of included studies

Study	Criterion 1	Criterion 2	Criterion 3	Criterion 4	Criterion 5	Criterion 6	Criterion 7	Criterion 8	Criterion 9	Criterion 10	Criterion 11	Total
Attene et al. [52]	0	1	0	1	0	0	0	1	0	1	1	5
Ramírez-Campillo et al. [39]	0	1	1	1	1	1	1	1	0	1	1	9
Ramírez-Campillo et al. [41]	0	1	0	1	0	0	0	1	0	1	1	5
Ramírez-Campillo et al. [40]	0	1	1	1	1	0	0	1	0	1	1	7
Campo et al. [42]	0	1	0	1	0	0	0	1	0	1	1	5
Chimera et al. [43]	0	1	0	1	0	0	0	1	0	1	1	5
Usman and Shenoy [44]	0	1	0	1	0	0	0	1	0	1	1	5
Irmischer et al. [45]	0	1	0	1	0	0	0	1	0	1	1	5
Kato et al. [46]	0	1	0	1	0	0	0	1	0	1	1	5
Makaruk et al. [47]	0	1	0	1	0	0	0	1	0	1	1	5
Martel et al. [48]	0	1	0	1	0	0	0	1	0	1	1	5
Ozbar [49]	0	0	0	1	0	0	0	1	0	1	1	4
Ozbar et al. [50]	0	1	0	1	0	0	0	1	0	1	1	5
Pereira et al. [18]	0	1	0	1	0	0	0	1	0	1	1	5
Rubley et al. [17]	0	0	0	1	0	0	0	1	0	1	1	4
Vescovi et al. [51]	0	1	0	1	0	0	0	1	0	1	1	5

Criterion 1 eligibility criteria were specified, *Criterion 2* subjects were randomly allocated to groups, *Criterion 3* allocation was concealed, *Criterion 4* the groups were similar at baseline regarding the most important prognostic indicators, *Criterion 5* there was blinding of all subjects, *Criterion 6* there was blinding of all therapists who administered the therapy, *Criterion 7* there was blinding of all assessors who measured at least one key outcome, *Criterion 8* measures of at least one key outcome were obtained from more than 85 % of the subjects initially allocated to groups, *Criterion 9* all subjects for whom outcome measures were available received the treatment or control condition as allocated, *Criterion 10* the results of between-group statistical comparisons are reported for at least one key outcome, *Criterion 11* the study provides both point measures and measures of variability for at least one key outcome

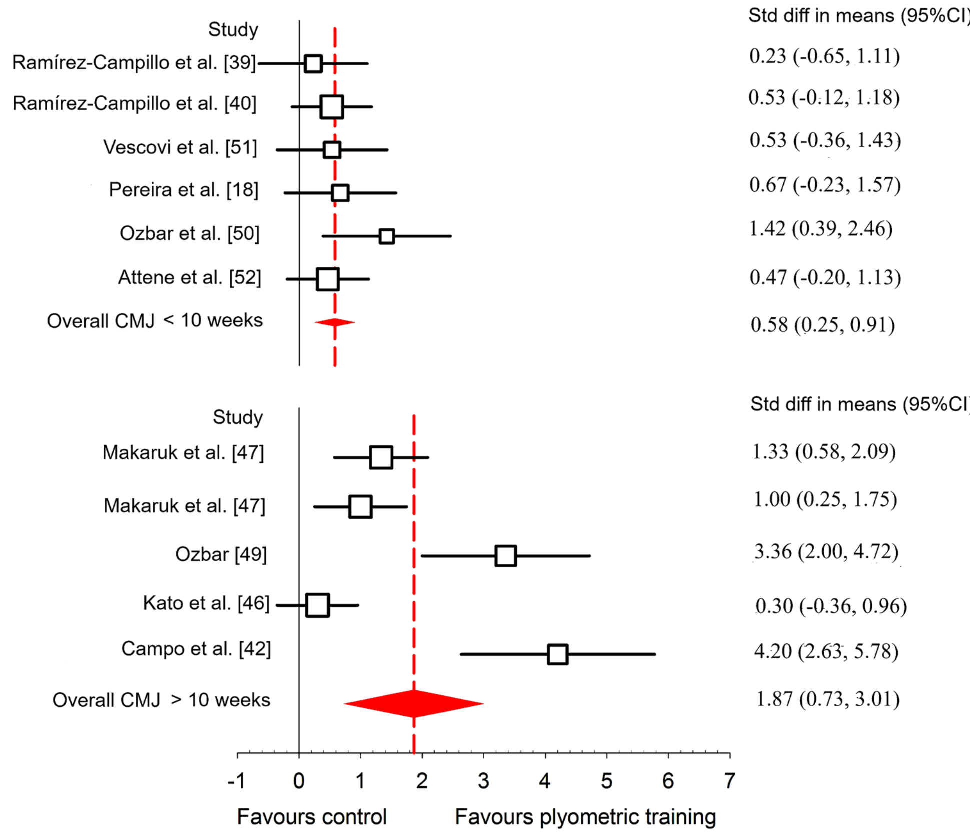


Fig. 2 Forest plot of the effect sizes and 95 % confidence intervals of the changes in countermovement jump performance. *CI* confidence interval, *CMJ* countermovement jump, *Std diff* standardised difference

showed that low-frequency plyometric training (1 day per week) has a positive effect (ES = 1.34 and 1.42) on VJ performance [17, 50]. The authors of the included studies [18, 44, 46–50, 52] recommend that plyometric training volume and intensity be periodically adjusted when prescribing short- and long-term plyometric training interventions for female athletes.

Our meta-analytic findings are in agreement with previous studies that confirmed the benefits of plyometric training on VJ performance in females [17, 18, 39–42, 44, 46–50, 52]. The varied effects of plyometric training on VJ performance of the various jump patterns (CMJ, SJ and DJ) were expected. It has been suggested that plyometric training is more effective at improving performance in jumps utilising the SSC (CMJ and DJ), as it enhances the elastic properties of the musculo-tendon unit and neural sequencing and firing rates of the involved motor units [26, 53–55]. The effect of plyometric training on VJ performance also varied between fast (DJ) and slow (CMJ) SSC movements. The results indicate that plyometric training is more effective for improving VJ performance in fast SSC movements [55]. The CMJ is a slow SSC movement, hence the moderate effect of plyometric training (ES = 1.09) on CMJ performance with hands-on-hips, and large effect (ES = 1.31) on CMJ performance with arm swing. Arm swing increases vertical ground reaction force causing an increase in jump height [56]. However, the observed effect differences of plyometric training on the CMJ with and without arm swing should be interpreted with caution, as one study reported unrealistic large gains (ES = 5.10) in CMJ performance with arm swing [44].

During the DJ, an athlete drops down from a specified height (30–90 cm), decelerates her body mass (eccentric phase) and rapidly transitions (amortisation phase) from the eccentric to concentric phase and subsequent flight phase. As a result, large eccentric loads and forces are experienced and developed during the DJ; similar eccentric forces are experienced during plyometric training, hence the observed very large positive effects (ES = 3.59) of plyometric training on DJ performance. Based on the mechanisms governing the respective jump movement patterns, it is expected that larger positive plyometric training effects would occur in the DJ and CMJ than in the SJ. The concentric-only SJ is arguably a non-plyometric ballistic exercise [57], due to the fact that during the SJ, that athlete pauses for 3–5 s in the bottom position (amortisation phase) following the eccentric phase prior to the concentric phase and subsequent flight phase. Pausing during the amortisation phase leads to the dissipation of elastic potential energy and in turn reduces the positive effects of the SSC, hence the small plyometric training

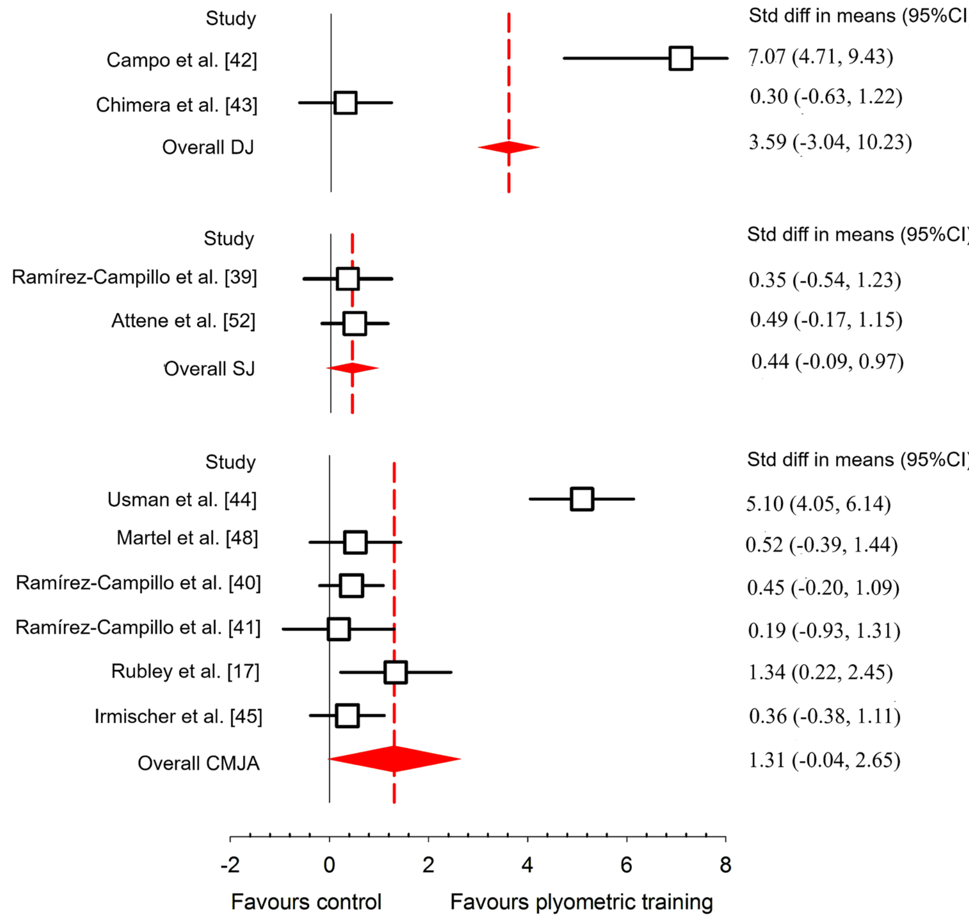


Fig. 3 Forest plot of the effect sizes and 95 % confidence intervals of the changes in drop jump, squat jump and countermovement jump with arm swing performance. *CI* confidence interval, *CMJA* countermovement jump with arm swing, *DJ* drop jump, *Std diff* standardised difference, *SJ* squat jump

effect (ES = 0.44) observed on SJ height performance in female athletes. The laws of training specificity were evident based on these findings, as the movement patterns performed during plyometric training also produced the largest improvements in performance.

An athlete's training status and experience may also influence the plyometric training effect. Untrained and amateur athletes will most likely have a greater response, and in turn a larger increase, in jump performance than trained and professional athletes. Fifteen of the included studies were conducted on collegiate and high school female athletes; therefore, a meta-analysis comparing sub-elite and elite female athletes was not conducted. Only one study [41] examined effects of plyometric training on elite athletes. Results of this study revealed a trivial plyometric training effect (ES = 0.19) on jump performance in elite female runners. The effect of plyometric training should also consider the athlete's level of biological maturation and chronological age. The benefits of plyometric training on CMJ with arm swing were greater in younger soccer players (ES = 1.34; mean age = 13.4 ± 0.5 years) [17] than in collegiate soccer players (ES = 0.45; mean age = 22.4 ± 2.4 years) [40]. This may also be a result of training status and baseline jump performance, as the collegiate athletes demonstrated higher baseline jump performance than the adolescent athletes.

Some limitations of this systematic review must be outlined. There were limitations in external validity, as most of the participants included were collegiate athletes and, therefore, no comparison could be made between elite, sub-elite and collegiate athletes. In addition, it was not feasible to use chronological age as a moderator variable, as only four studies included young athletes. Another potential limitation was the low ('fair') methodological quality ratings of the majority of studies (14 of 16 studies) included in this review. The limited number of studies also make it difficult to draw any definitive conclusions. It is recommended that future research investigating the effects of plyometric training on jump performance improve the quality of their study designs.

5 Conclusion

The present meta-analysis and systematic review demonstrates that plyometric training improves VJ performance in female athletes regardless of age, type of sport and competition level. However, the effectiveness of plyometric training on jump performance is jump pattern specific, as the following plyometric training effects were observed: very large increases in DJ performance, large increases in CMJ performance and small increases in SJ performance. Training duration also appears to influence the effectiveness of plyometric training, as longer training durations ($\geq$10 weeks) provide larger improvements in jump performance. In closing, plyometric training can be implemented as an effective form of training to increase VJ performance in female athletes.

Compliance with Ethical Standards

Funding No sources of funding were used to assist in the preparation of this review.

Conflict of interest Emilija Stojanović, Vladimir Ristić, Daniel Travis McMaster and Zoran Milanović declare that they have no conflicts of interest relevant to the content of this review.

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